Jiawei Zhao **Last update: March 16, 2026**

CONTACT INFORMATION

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PROFESSIONAL SUMMARY

- Data Scientist and Machine Learning Engineer with 5+ years of experience developing data-driven analytical systems across research and operational environments. Experienced in predictive modelling, anomaly detection, and early-warning analytics to support risk monitoring and decision-making in public sector contexts.
- Strong background in building and deploying production machine learning pipelines, including model deployment, monitoring, and data quality frameworks. Skilled in translating complex analytical insights into practical solutions for operational teams and stakeholders.
- Experienced working with large-scale data platforms, statistical modelling, and modern data science workflows using Python, SQL, and cloud-based systems. Experience working in multi-disciplinary environments translating analytical models into operational decision-support tools.

PROFESSIONAL EXPERIENCE

- **Machine Learning Engineer & Data Scientist, New Zealand Institute for Public Health and Forensic Science (PHFScience NZ)** Nov 2022 – Present
  - Led the design and deployment of machine learning and advanced analytics systems supporting organisational decision-making and operational analytics.
  - Developed predictive and early-warning analytics models for public health risk detection and operational monitoring, applying statistical modelling, anomaly detection, and forecasting techniques to support real-time situational awareness.
  - Designed and maintained production machine learning pipelines including model deployment, monitoring, and performance evaluation to ensure reliable operational analytics.
  - Implemented data quality frameworks and MLOps practices to ensure transparency, reliability, and maintainability of analytical systems.
  - Collaborated with multidisciplinary stakeholders to translate analytical findings into actionable insights supporting evidence-based decision-making.
  - Contributed to national research and innovation programmes by preparing technical proposals and research documentation for government funding initiatives.
- **Data Scientist, The Mindlab Inc, NZ** March 2022 – November 2022
  - Developed data analytics workflows and dashboards to support organisational reporting and operational decision-making.
  - Worked with engineering and product teams to implement data quality controls and maintain reliable data processing pipelines.
  - Built and maintained data pipelines using Google Cloud, Firebase, SQL databases, and Linux environments for data integration and transformation.
  - Applied statistical and machine learning methods for trend analysis and predictive modelling, providing insights to support strategic planning and data-driven decision making.
- **Data Scientist PlantTech Inc, NZ** Jan 2022 – March 2022
  - Developed machine learning models for large-scale agricultural sensor data, applying deep learning and image analysis techniques to support crop monitoring and automated quality assessment.
  - Designed predictive analytics algorithms for equipment performance monitoring and maintenance forecasting, improving operational efficiency in agricultural production systems.

  - Built data processing pipelines integrating multi-sensor datasets and computer vision models to detect and classify kiwi fruit conditions across production environments.
  - Collaborated with research scientists and engineering teams to translate analytical requirements into practical machine learning solutions and actionable insights.

- **AI Research Intern, Alibaba Inc, China** September 2020 – March 2021
  - Developed natural language processing and speech processing models using transformer-based architectures to support multilingual speech translation and language understanding tasks.
  - Applied machine learning techniques including model compression and knowledge distillation to improve computational efficiency and training performance of large neural models.
  - Contributed to the research and development of multilingual speech translation systems, resulting in a peer-reviewed publication at EMNLP 2021.
  - Collaborated with engineering teams on system integration, model evaluation, and performance testing, producing technical documentation and validation reports for production readiness.

- **Visiting AI Researcher, IXA Group, University of the Basque Country, Spain** March 2020 – July 2020
  - Conducted research on machine learning approaches for natural language processing in low-resource language settings.
  - Developed algorithms for analysing multilingual textual data and extracting features from high-dimensional datasets.
  - Investigated cross-lingual word embedding techniques to address data scarcity and improve model performance across languages.
  - Produced technical reports and research documentation summarising analytical findings and experimental results.

Technical Skills

- **Programming:** Python (5+ years), SQL, Scala, R, Java
- **Machine Learning:** Scikit-Learn, PyTorch, forecasting, classification, clustering, anomaly detection, statistical modelling
- **Data Platforms:** SQL databases, data pipelines, cloud platforms (Azure)
- **Tools:** Git (GitHub, GitLab), PowerBI, HuggingFace
- **Systems:** Linux/Unix (bash scripting)

Education

Ph.D. in Artificial Intelligence / Machine Learning
Massey University, Auckland, New Zealand — Apr 2017 – Jun 2022
Visiting Ph.D. Researcher
University of the Basque Country (IXA Group), Spain — Mar 2020 – Jul 2020
M.Sc. in Computer Science
Hebei University of Technology, Tianjin, China — Sep 2014 – Mar 2017
B.Sc. in Computer Science
Hebei University of Technology, Tianjin, China — Sep 2010 – Jun 2014

Selected Publications

1. Zhao, J. et al. "Mutual-Learning Improves End-to-End Speech Translation." *Empirical Methods in Natural Language Processing (EMNLP)*, 2021.

2. Zhao, J. et al. "Non-Linearity in Mapping Based Cross-Lingual Word Embeddings." *Language Resources and Evaluation Conference (LREC)*, 2020.

Honours & Awards

- SNCS International Student Scholarship, Massey University 2017 – 2020
- Visiting Research Scholar, Massey University 2020
- Key Student Scholarship, Hebei University of Technology 2016

Professional Activities

- Reviewer, Empirical Methods in Natural Language Processing (EMNLP)
- Reviewer, ACM Transactions on Asian and Low-Resource Language Information Processing
- Assistant Reviewer, Image and Vision Computing New Zealand (IVCNZ)

Research Activities

- Assisting and attending *The 2021 Conference on Empirical Methods in Natural Language Processing (EMNLP)* 2021
- Invited research visit to the University of the Basque Country in Spain. 2020